DOCUMENT REGISTRY ID: GEO-RS-119

Regional Geological Assessment â€” North Karanpura

Basin Tectonics, Stratigraphic Sequence, CBM Survey & UNFC Inventory

CATEGORY	MINE / BLOCK	YEAR	CLASSIFICATION
Geological	North Karanpura	2023	RESTRICTED / VERIFIED

DOCUMENT VERIFICATION & ACCESS REGISTER

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Security Status: Verified & SHA-256 Checksum Certified

Total Document Size: 24 Certified Multi-Page Sections

Standard Operating Procedure: SOP-CI-MIN-2025-A

System Node: CLOUD-KNOWLEDGE-NODE-EAST-01

Storage Archive Date: 2025-02-14 09:30:00 UTC

SECTION 1.0 â€” DOCUMENT INFORMATION & ASSESSMENT SCOPE

Official Technical Record & Geological Data Logging

The Regional Geological Assessment of the North Karanpura Coalfield synthesizes regional basin tectonics, stratigraphy, structural fault mapping, and coal quality trends.

Data includes deep borehole correlations, seismic reflection profiles, and proximate chemical analyses across 1,230 square kilometers of the Gondwana rift basin.

Regional Assessment Metadata

Parameter	Specification / Details	Authority / Agency	Status
Basin Name	North Karanpura Coal Basin	Geological Survey of India	Standardized
Total Basin Area	1,230 sq. kilometers	State Directorate of Geology	Mapped
Stratigraphic Focus	Barakar & Karharbari Formations	CMPDI Exploration Wing	Verified
Estimated Resource	14.24 Billion Tonnes	UNFC National Coal Inventory	Gazetted

SECTION 2.0 â€” EXECUTIVE SUMMARY & BASIN POTENTIAL

Official Technical Record & Geological Data Logging

North Karanpura represents one of the largest coal resource repositories in Eastern India, with prime multi-seam opencast blocks suited for large-scale mechanization.

Over 40 coal seams have been identified, with composite seam thicknesses exceeding 35 meters in the central depression.

CRITICAL TECHNICAL OBSERVATION & DIRECTIVE:

The basin contains significant Coalbed Methane (CBM) potential estimated at 32.4 Billion Cubic Meters in deep coal horizons (> 400m depth).

VERIFIED PARAMETER REGISTRATION:

â€¢ **Total Geological Resource:**	14.24 Billion Tonnes	Proved: 6.8 BT, Indicated: 7.4 BT
â€¢ **Average Coal Thickness:**	28.5 meters	Cumulative seam thickness
â€¢ **Dominant Coal Rank:**	High Volatile Bituminous (G14 to G7)	Thermal Grade

SECTION 3.0 â€” REGIONAL PHYSIOGRAPHY & DRAINAGE PATTERN

Official Technical Record & Geological Data Logging

The North Karanpura Basin forms an elliptical east-west elongated depression drained by the Damodar River and its tributaries (Garhi, Haharo, and Chhati streams).

Average surface elevation ranges from 380m MSL in valley floors to 620m MSL on surrounding Precambrian granite gneiss plateaus.

SECTION 4.0 â€” REGIONAL TECTONIC FRAMEWORK OF DAMODAR VALLEY

Official Technical Record & Geological Data Logging

The basin is bounded on the north by the Southern Boundary Fault of the Hazaribagh Plateau and on the south by the Ranchi Plateau Metamorphic Complex.

VERIFIED PARAMETER REGISTRATION:

â€¢ **Basin Structure:**	`Asymmetric Synclinal Trough`	Steep northern margin
â€¢ **Faulting Intensity:**	`Moderate to High`	Major E-W and NW-SE trends
â€¢ **Basement Depth:**	`Up to 1,200 meters`	In central sub-basin

SECTION 5.0 â€” BASIN EVOLUTION & SEDIMENTOLOGICAL HISTORY

Official Technical Record & Geological Data Logging

Sedimentation began during the Permo-Carboniferous with glacial Talchir boulder beds deposited on scoured Precambrian basement.

This was followed by extensive fluvio-deltaic braided river systems depositing the coal-bearing Barakar sandstones under temperate humid paleoclimates.

SECTION 6.0 â€” STRATIGRAPHIC CLASSIFICATION OF GONDWANA SEQUENCE

Official Technical Record & Geological Data Logging

The standard Gondwana stratigraphic succession recognized in North Karanpura is detailed below.

Stratigraphic Succession of North Karanpura Basin

Age / Period	Group / Formation	Approx Thickness (m)	Dominant Lithology
Lower Cretaceous	Mahadeva Formation	250 m	Red ferruginous sandstone & pebble conglomerate
Upper Permian	Raniganj Formation	450 m	Fine sandstone, carbonaceous shale, thin coal seams
Middle Permian	Barren Measures	380 m	Dark grey ironstone shales, non-coal bearing
Lower Permian	Barakar Formation	650 m	Coarse arkosic sandstone, major thick coal seams
Upper Carboniferous	Talchir Formation	120 m	Tillites, needle shales, khaki-green sandstones
Precambrian	Basement Complex	Unconformity	Granite gneiss, amphibolite, mica schist

SECTION 7.0 â€” TALCHIR FORMATION & BASAL GLACIAL TILLITES

Official Technical Record & Geological Data Logging

The Talchir formation crops out along the northwestern and southern fringes of the basin, resting unconformably on crystalline basement rocks.

SECTION 8.0 â€” BARAKAR FORMATION DEPOSITIONAL FACIES & COAL CYCLES

Official Technical Record & Geological Data Logging

The Barakar Formation is the primary economic target, housing 8 major coal seam horizons numbered from Seam I (base) to Seam VIII (top).

Barakar Coal Seam Group Nomenclature

Seam Group	Stratigraphic Interval (m)	Avg Seam Thk (m)	Coal Quality Grade
Upper Barakar (Seams VI-VIII)	Depth 50 - 220 m	12.4 m	Grade G-4 to G-5
Middle Barakar (Seams IV-V)	Depth 220 - 450 m	14.8 m	Grade G-5 to G-6
Lower Barakar (Seams I-III)	Depth 450 - 650 m	8.2 m	Grade G-3 to G-4

SECTION 9.0 â€” BARREN MEASURES & RANIGANJ FORMATIONS

Official Technical Record & Geological Data Logging

The Barren Measures Formation consists of 350-400m of monotonous dark shales with sideritic ironstone bands, acting as a regional aquitard.

SECTION 10.0 â€” MAJOR FAULT SYSTEMS & STRUCTURAL FAULT BLOCKS

Official Technical Record & Geological Data Logging

The basin is dissected into five major fault blocks by gravity faults with throws ranging from 20m to 180m.

Major Regional Fault Systems

Fault Name	Strike Direction	Dip & Throw (m)	Structural Influence
Tandwa Boundary Fault	N 80 W	75 N / Throw 180 m	Basin Boundary Definer
Garhi River Fault	N 45 W	65 SW / Throw 65 m	Seam Discontinuity Zone
Pipwar Transverse Fault	N 15 E	80 SE / Throw 45 m	Block Boundary

SECTION 11.0 â€” COAL SEAM GEOMETRY & SPLITTING CHARACTERISTICS

Official Technical Record & Geological Data Logging

A notable geological feature in North Karanpura is extensive seam splitting towards the east, where the composite Dakra-Bukbuka seam splits into four distinct leaves.

SECTION 12.0 â€” COAL PETROGRAPHY & MACERAL COMPOSITION

Official Technical Record & Geological Data Logging

Maceral analysis shows dominance of Vitrinite (48-58%) and Inertinite (32-42%) with low Liptinite (4-8%), typical of Indian Permian coals.

Petrographic Maceral Analysis (%)

Seam Group	Vitrinite %	Inertinite %	Liptinite %	Mineral Matter %
Upper Barakar	54.2%	34.8%	5.2%	5.8%
Middle Barakar	51.0%	37.5%	4.8%	6.7%
Lower Barakar	56.8%	32.1%	6.1%	5.0%

SECTION 13.0 â€” VITRINITE REFLECTANCE & COAL RANK EVOLUTION

Official Technical Record & Geological Data Logging

Mean random vitrinite reflectance (Ro max) ranges from 0.65% to 0.88%, indicating high-volatile bituminous rank.

VERIFIED PARAMETER REGISTRATION:

â€¢ Mean Vitrinite Reflectance (Ro):	0.82%	Bituminous rank
â€¢ Volatile Matter (dmmf):	34.5%	High volatile coal
â€¢ Hydrogen Content:	4.8%	Favorable for gasification

SECTION 14.0 â€” COALBED METHANE (CBM) RESOURCE ASSESSMENT

Official Technical Record & Geological Data Logging

Gas desorption tests on deep cores indicate in-situ methane content of 4.5 to 8.2 m3 per tonne of coal at depths below 450 meters.

Deep Seam CBM Gas Content Survey

Block / Sector	Depth Range (m)	Gas Content (m3/T)	Methane Purity %
Tandwa Deep Sector	450 - 750 m	7.4 m3/T	94.8% CH4
Chano-Rikba Sector	500 - 800 m	8.2 m3/T	96.2% CH4
Pachra Deep Sector	400 - 650 m	5.8 m3/T	93.5% CH4

SECTION 15.0 â€” DEEP EXPLORATION DRILLING & BASIN CORRELATION

Official Technical Record & Geological Data Logging

A network of 48 regional deep exploratory boreholes has validated seam continuity across deep synclinal troughs.

SECTION 16.0 â€” GEOPHYSICAL SEISMIC & GRAVITY ANOMALY PROFILES

Official Technical Record & Geological Data Logging

2D seismic reflection profiles over 140 line-kilometers confirm basement depression architecture and fault block compartmentalization.

VERIFIED PARAMETER REGISTRATION:

â€¢ Bouguer Gravity Anomaly:	`-35 to -65 mGal`	Reflecting 1,200m sedimentary fill
â€¢ Seismic P-Wave Velocity (Barakar):	`3,200 - 3,800 m/s`	Sandstone intervals
â€¢ Coal Seam P-Wave Velocity:	`2,100 - 2,400 m/s`	High acoustic contrast

SECTION 17.0 â€” HYDROGEOLOGICAL REGIME & BASIN GROUNDWATER DYNAMICS

Official Technical Record & Geological Data Logging

Regional groundwater flows radially towards the Damodar River channel. Recharge occurs primarily through permeable Barakar sandstone outcrops along the basin margins.

SECTION 18.0 â€” GEOTHERMAL GRADIENT & IN-SITU STRESS REGIME

Official Technical Record & Geological Data Logging

Geothermal gradient is established at 28.5 deg C per kilometer depth. Maximum horizontal compressive stress trends N 75 E.

VERIFIED PARAMETER REGISTRATION:

â€¢ Geothermal Gradient:	28.5 deg C / km	Normal intracratonic gradient
â€¢ Virgin Rock Temp at 600m:	44.2 deg C	Manageable for deep ventilation
â€¢ Horizontal Stress Ratio (k):	1.45	Compressive stress regime

SECTION 19.0 â€” GEOLOGICAL RESOURCE CALCULATION & UNFC CATEGORIZATIO

Official Technical Record & Geological Data Logging

Resources are categorized under the United Nations Framework Classification (UNFC) System based on borehole grid density.

North Karanpura Coal Resource Inventory by UNFC Class

UNFC Category	Grid Spacing	Resource (Billion Tonnes)	Reliability Index
Class 111 (Proved)	< 400 m grid	6.84 BT	High Confidence
Class 122 (Indicated)	400 - 1000 m grid	5.12 BT	Moderate Confidence
Class 333 (Inferred)	> 1000 m grid	2.28 BT	Preliminary
TOTAL BASIN RESOURCE	All Sectors	14.24 BT	National Inventory

SECTION 20.0 â€” BASIN-SCALE EXTRACTION STRATEGY & MEGA PITS

Official Technical Record & Geological Data Logging

Over 60% of proved resources are amenable to mega-opencast extraction with pit capacities designed between 15 MTPA and 30 MTPA.

SECTION 21.0 â€” GEOTECHNICAL ROCK MASS RATING (RMR) FOR MINING

Official Technical Record & Geological Data Logging

Bieniawski Rock Mass Rating (RMR) for Barakar sandstones ranges from 58 to 68 (Fair to Good Rock), supporting stable 70 deg highwall bench angles.

VERIFIED PARAMETER REGISTRATION:

â€¢ Roof Sandstone RMR:	64 (Good Rock)	Cohesive and competent
â€¢ Coal Seam RMR:	52 (Fair Rock)	Regular cleat development
â€¢ Floor Shale RMR:	46 (Fair Rock)	Requires dry sump maintenance

SECTION 22.0 â€” ENVIRONMENTAL GEOLOGY & MINE WATER QUALITY

Official Technical Record & Geological Data Logging

Low pyritic sulphur content (< 0.5%) in coal and surrounding sediments prevents Acid Mine Drainage (AMD) formation across all basin sectors.

SECTION 23.0 â€” CONCLUSIONS, REGIONAL OUTLOOK & SIGN-OFF

Official Technical Record & Geological Data Logging

The North Karanpura Coalfield represents a vital long-term pillar for national energy security, possessing multi-billion tonne opencast reserves.

Digitally validated and archived in Coal Intelligence Enterprise Repository.

CRITICAL TECHNICAL OBSERVATION & DIRECTIVE:

Official Geological Assessment Document ID: GEO-RS-119.

VERIFIED PARAMETER REGISTRATION:

â€¢ **Chief Regional Geologist:**	Dr. K. N. Rao, Ph.D (Geology)	Certified on 04-Sep-2023
â€¢ **Director of Exploration:**	Sunita Mahato, M.Sc	Approved on 06-Sep-2023